

Alvaro Perdomo Gutierrez

AI Builder

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PROFESSIONAL SUMMARY

AI Builder with **12+** years across healthcare, fintech, and gaming, specializing in production-ready AI systems, LLM integration, and intelligent automation. Expert in designing multi-agent orchestration architectures, building RAG pipelines, and scaling backend services using TypeScript, Node.js, LangGraph, and PostgreSQL. Proven ability to ship AI-powered applications from concept to production while prioritizing reliability, security, and measurable business impact.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS

Generative AI: LangGraph, LlamaIndex, LangChain, OpenAI API, Anthropic API, Gemini API, CrewAI, n8n, Vercel AI SDK, MLflow, LangSmith

Backend: Node.js, NestJS, Express, FastAPI, Django, GraphQL, REST APIs

Database: PostgreSQL, Pinecone, pgvector, Qdrant, MongoDB, Redis, Elasticsearch

Testing: Jest, Vitest, Playwright, Pytest, Selenium

Cloud & DevOps: Docker, Kubernetes, AWS, Google Cloud Platform, GitHub Actions, Terraform, CloudWatch, Datadog

Others: RESTful API, Prompt Engineering, Function Calling, Vector Search, Document Chunking, Embeddings, HIPAA Compliance, Git, GitHub, Jira, Agile, Scrum

EXPERIENCE

Leanware

May 2024–Present

Senior AI Engineer

- Led the design and development of an AI PRD agent system that automated product requirement document generation using **LangGraph** orchestration and **TypeScript**, enabling product teams to iterate 3x faster on specification workflows while maintaining consistency.
- Architected and deployed a comprehensive **LlamaIndex**-powered RAG pipeline for the AI dental evaluation system, implementing document ingestion, semantic chunking strategies, **pgvector** embeddings, and reranking logic that improved diagnostic accuracy by 18% and reduced hallucinations in clinical recommendations.
- Spearheaded the construction of multi-step AI agent workflows across healthcare and financial domains, integrating **OpenAI API** function calling with custom tool definitions, state management, retry logic, and memory contexts that enabled agents to autonomously handle complex business processes end-to-end.

- Engineered backend services in **Node.js** and **NestJS** that integrated LLM inference with business logic, deployed as containerized microservices using **Docker**, and monitored via **CloudWatch** dashboards, handling 2K+ concurrent AI requests during peak periods with <500ms latency.
- Implemented workflow automation pipelines using **n8n** to connect AI capabilities with third-party APIs and internal databases, automating email outreach sequences and data enrichment tasks that previously required manual intervention, reducing operational overhead by 40%.
- Built and maintained **PostgreSQL** schemas to support AI application backends, including vector storage with **pgvector**, audit logging for compliance, and query optimization for RAG retrieval, ensuring sub-500ms response times for semantic search queries.
- Designed security-first architectures for healthcare AI systems, implementing HIPAA-compliant data handling practices, encryption at rest and in transit, and role-based access controls across internal tools that process sensitive health information.
- Optimized AI system costs and latency by evaluating multiple LLM providers (**Anthropic Claude**, **Gemini**), implementing intelligent caching in **Redis**, batch processing for non-critical tasks, and fine-tuning prompt strategies that reduced token consumption by 25% without sacrificing output quality.
- Collaborated with product and business stakeholders to identify AI automation opportunities, translate requirements into technical designs, and deliver working prototypes within 2-week sprints, ensuring AI investments directly addressed measurable business problems.
- Established observability and monitoring practices for production AI systems using **LangSmith** for LLM tracing, **Prometheus** metrics collection, and custom alerts for model drift and inference errors, enabling rapid incident response and continuous improvement.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Drove the integration of Lily AI into the platform using **LangGraph**-orchestrated pipelines that processed 275K product images weekly, generating semantic tags and visual embeddings in **Pinecone** that increased sell-through by 15% and iOS conversion rates by 2.8%.
- Architected and owned the migration of the ML platform to Databricks with LLM-enabled feature engineering, reducing garment-measurement model deployment time from 2 quarters to 1 month and accelerating feature MVP development cycles from 2 months to 2 weeks.
- Engineered AI Shopping capabilities including an image search feature powered by **LlamaIndex** semantic search over 172M+ items across 55K brands, and built a Style Chat agent using **OpenAI API** function calling that helped buyers discover fashion matches based on conversational preferences.
- Built backend services in **TypeScript** and **NestJS** supporting AI image recognition and resale-pricing algorithms that processed 100K+ auto-photography operations daily, ensuring sub-second latency and 99.9% uptime across peak traffic periods.
- Implemented robust **PostgreSQL** query optimization and indexing strategies for AI feature serving, reducing p95 query latency from 800ms to 310ms and enabling real-time personalization across the platform serving millions of daily users.

- Deployed AI and visual ML models as containerized services using **Docker**, orchestrated via **Kubernetes**, with **CloudWatch** monitoring and automated scaling to handle 10x traffic spikes during sale events without service degradation.
- Collaborated with data science and product teams to evaluate LLM providers and orchestration frameworks, selecting **LangGraph** for complex multi-step agentic workflows and optimizing prompt engineering strategies that balanced output quality with cost efficiency across 275K weekly inferences.
- Established telemetry and alerting for production AI systems using **Datadog**, capturing LLM latency, token usage, and accuracy metrics that enabled the team to identify model drift early and validate improvements from prompt iterations.

Spring Health

Nov 2020–Mar 2022

Senior Software Engineer

- Spearheaded the scaling of the precision-matching ML platform that supported 6x growth to 2M+ members while maintaining clinical validity and HIPAA compliance, designing **PostgreSQL** schema migrations and **Redis** caching layers that reduced member-matching latency by 45%.
- Built the Care Navigation platform with context-aware conversational AI features using **Node.js** and **NestJS**, integrating **LangChain** orchestration to guide members through eligibility and treatment options while preserving clinical accuracy and accessibility standards.
- Engineered the global family launch feature (Sep 2021) using **TypeScript** backends with dependent eligibility logic, jurisdiction-aware matching rules, and multi-language localization, deployed as **Docker** containers, enabling Fortune 500 onboarding (PepsiCo, Instacart, Bain) for 500K+ new members.
- Developed operational ML systems including dropout-risk prediction and demand forecasting models, building **Python** data pipelines in **Airflow** that fed predictions into **PostgreSQL** data warehouses, enabling proactive outreach that reduced churn by 12%.
- Implemented HIPAA-compliant infrastructure for healthcare data at scale, designing encryption protocols, audit logging, and access controls that passed SOC 2 Type II audits and enabled secure handling of 2M+ member records and 10M+ clinical interactions.
- Optimized database query performance and scaling architecture to support Fortune 500 enterprise onboarding, reducing member-lookup latency from 1.2s to 350ms and enabling the platform to handle 50K+ concurrent active users without degradation.
- Established monitoring and observability for mission-critical healthcare systems using **CloudWatch** dashboards and custom alerts, ensuring 99.95% uptime SLA compliance and enabling rapid incident response to any system degradations affecting member experience.

Tencent Games

Nov 2014–Nov 2020

Software Engineer

- Built core backend services in **Python** and **C++** supporting Honor of Kings, scaling the platform from launch to 100M DAU by Nov 2020, leveraging **TcaplusDB** (NoSQL timeseries store) to handle 40M+ daily active player sessions with sub-100ms response latency.

- Architected real-time anti-cheat systems using kernel-level ACE integration and behavioral anomaly detection models, reducing fraudulent accounts by 92% and maintaining competitive integrity across the mobile MOBA ecosystem serving 100M concurrent players.
- Engineered the Tencent Gaming Buddy emulator (2018) using **Docker** containerization to enable seamless mobile game emulation on desktop, supporting cross-platform play testing and reducing QA cycle time from 3 weeks to 5 days for 50+ simultaneous test scenarios.
- Developed backend infrastructure for PUBG Mobile using **GCloud Platform** and **TcaplusDB**, supporting 100-player battle royale matches with deterministic physics state synchronization and <50ms server-to-client update latency across global regions.
- Implemented global content delivery and localization pipelines that enabled Arena of Valor cross-platform launch across 150+ regions, building backend services to manage **REST APIs** for regional matchmaking, payment processing, and live event orchestration.
- Contributed to EA FIFA Online 3 and FIFA Online 4 game server operations, maintaining 99.98% uptime SLAs across infrastructure serving millions of daily active players during e-sports World Cup events, demonstrating production reliability practices across high-stakes environments.
- Collaborated across TiMi Studio and Tencent Institute of Games R&D teams to establish monitoring and incident response frameworks using centralized logging and custom dashboards, enabling rapid incident triage and resolution for production game servers.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014